Mobile L4S

draft-sporeba-tsvwg-mobile-l4s

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Problem statement and scoping

- Deploying L4S in a mobile ecosystem requires co-operation across multiple layers: the network stack, the host operating system, link-layer drivers and firmware (e.g., Wi-Fi and cellular modem), and the network.
- The draft aims to provide operational guidelines (i.e. recommended thresholds) and reiterates the relevant parts of RFCs 9330-9332.

System recommendations

- Detect packet filtering and errors, remember them to avoid future latency impact
    - Draft recommends using probes to a known L4S-enabled destination, to remember the state of per-network support
    - Draft proposes caching per-IP or host support if viable
    - Proposed cache TTL is 7 days

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Link-layer subsystems recommendations

- While mixing L4S traffic with other traffic is allowed by 9331 §5.4.1 under certain conditions, draft outlines the consequences of mixing and recommends having a strictly distinct L4S queue
  - RFC 9332 outlines the L4S and Classic queue distinction.
  - Some modem subsystems already have two queues: high and low priority.
- Packets should not be reordered (on transmit and receive path)
  - A problem that draft-white-intarea-reordering also describes and proposes to solve

Network recommendations

- Avoid non-compliant prioritisation
  - Some networks would like to allocate a slice of traffic to ECN marked traffic (good)
  - without implementing any ECN marking or low-latency-queues (not so good)
  - We expect apps to mark ECT(1) without responding to ECN markings, just to get in the new slice.
- Intentionally tried to minimize the overlap with draft-livingood-low-latency-deployment

Questions / comments

- Does the scope of the document sound right?
  - Extend, e.g. to cover the relevant mobile network components or apps?
  - Shrink, to avoid overlap with other documents?
- Is this of interest to TSV?